MIL-C-13600A
14 MAY 1962
SUPERSEDING
MIL-C-13600
25 AUGUST 1954

MILITARY SPECIFICATION

CUPRIC OXIDE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one grade of cupric oxide intended for Ordnance applications.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR–S–366 — Sieves; Standard for Testing Purpose.

UU–S–48 — Sacks, Paper, Shipping.

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–109 — Inspection Terms and Definitions.

MIL–STD–129 — Marking for Shipment and Storage.

PUBLICATIONS

ORDNANCE CORPS

ORD–M608–11 — Procedures and Tables for Continuous Sampling by Attributes.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE PUBLICATION

Consolidated Freight Classification Ratings, Rules, and Regulations.

FSC 6810

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:54Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification, Rules and Container Regulations.

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth Street, N.W., Washington 6, D.C.)

3. REQUIREMENTS

3.1 Moisture. The moisture content shall be not more than 0.2 percent when tested as described in 4.3.1.

3.2 Cupric Oxide. The cupric oxide content shall be not less than 95.0 percent when tested as described in 4.3.2.

3.3 Acid insoluble matter. The acid insoluble matter shall be not more than 0.5 percent when tested as described in 4.3.3.

3.4 Granulation. The granulation of cupric oxide shall be such that not less than 95 percent through a No. 200 U.S. Standard Sieve when tested as described in 4.3.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

4.1.1 *Contractor quality assurance system.* If the contractor desires to utilize a quality assurance system, which is at variance with the quality assurance provisions of 4.2.2 and 4.3 and other documents referenced herein, he shall submit a written description of the system to the contracting officer for approval prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures methods of inspections (measuring and testing equipment), and provisions for control and disposition of nonconforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance provisions required by the provisions of 4.2.2 and 4.3 and the other documents referenced herein. The contractor shall not be restricted to the inspection station or the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any changes to the written procedure that effects the degree of assurance required by this specification or other documents referenced herein.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product submitted:

(a) A statement that the lot complies with all quality assurance pro-

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:54Z
Check the source to verify that this is the current version before use.

visions of the approved current written description of the system.

(b) Quantity of product inspected.

(c) Results obtained for all inspection performed.

(d) Specification number and date, together with an identification and date of changes.

(e) Certificates of analysis on all material procured directly by the contractor when such material is controlled by Government specifications listed in any of the contractual documents.

(f) Quantity of product in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance procedure (see 4.1.1), this detail specification, and other contractual documents as a guide, the Government inspector shall verify all quality assurance operations performed by the contractor. Verification shall be in accordance with a or b as applicable, the decision being the responsibility of the procuring activity. In either case, the inspector shall also ascertain, prior to acceptance, that all quality assurance provisions of other specifications referenced in any of the contractual documents have been complied with. Deviations from prescribed or agreed upon procedures discovered by the Government inspector shall be brought to the attention of the supplier. Disposition of the product and remedial action shall be as directed by the Government inspector and, depending on the nature of the deviation, may consist of lot rejection, screening, resampling, reinstruction of the supplier's employees, or other appropriate action:

(a) Verification at the point of manufacture shall be accomplished at unscheduled intervals in accordance with 4.1.3.1 and 4.1.3.2.

(b) Verification at the point of delivery shall be in accordance with 4.1.3.2.

4.1.3.1 *Surveillance.* Surveillance shall include, but is not limited to:

(a) Observation of procedures concerning lot formation and identification.

(b) Observation of sampling procedures and application of acceptance criteria.

(c) Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification or approved equivalents thereto.

(d) Review of procedures for control and disposition of nonconforming material.

4.1.3.2 *Product inspection.* Product inspection shall consist of Government inspection of product which has been previously inspected by the contractor and found to meet the quality assurance provisions of this specification. The inspection by the Government shall be performed in order to determine that the product is of the quality required by this specification and that the contractor's records are reliable.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:54Z
Check the source to verify that this is the current version before use.

one or more batches of cupric oxide, produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of cupric oxide that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogenous.

4.2.2 *Examination.* Sampling plans and procedures for the following classification of defects shall be in accordance with Standard MIL–STD–105. Continuous sampling plans, in accordance with Handbook ORD–M608–11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect.

4.2.2.1 *Paper shipping sack, after filling (see 5.1).*

Categories	*Defects*	*Method of inspection*
Critical: None defined.		
Major:	*AQL 0.25 percent*	
101.	Sack torn, cut or punctured	Visual
102.	Closure incomplete or improper	Visual
Minor:	*AQL 1.00 percent*	
201.	Marking misleading or unidentifiable	Visual
202.	Evidence of moisture	Visual-Tactile

4.2.3 *Testing.*

4.2.3.1 *Sampling by lot.* A random sample of 10 containers shall be selected from each lot. When lots are comprised of 10 containers or less, each container shall be sampled.

4.2.3.1 *Preparation of composite.* A 1.0 ± .2 ounce primary sample of cupric oxide shall be removed from each of the 10 containers in order to equal 10 ounces. If there are less than 10 containers, equal primary samples in sufficient quantity to equal 10 ounces, shall be removed from each container. The individual primary samples shall then be combined in order to form a homogeneous composite sample of 10 ounces and subjected to the tests specified in 4.3. If the composite sample fails to comply with any of the requirements specified, the lot shall be rejected.

4.3 Test methods and procedures. The following test methods shall be performed as follows:

4.3.1 *Determination of moisture.* Approximately 5 gram (gm) of the sample shall be transferred to a tared weighing bottle and accurately weighed. The sample shall be dried for 1 hour at 100° to 150° centigrade (C.). The bottle and sample shall be cooled in a desiccator and weighed. The loss in weight shall be calculated to percentage of moisture in the sample.

4.3.2 *Determination of cupric oxide.* Transfer an accurately weighed portion of approximately 1 gm. of the sample to a 200 milliliter (ml) beaker. Add a 50 ml portion of 10 percent by volume of sulfuric acid solution to a beaker. Heat the beaker and contents on a hot plate and allow the solution to boil for 10 minutes. Remove the beaker and contents from the hot plate and allow the solution to cool to room temperature. Filter the solution using a No. 40 Whatman filter paper (or equivalent) and catch the filtrate in a 200 ml tall form beaker. Wash the filter paper with three 25 ml portions of distilled water and catch the washings in the

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:54Z
Check the source to verify that this is the current version before use.

tall form beaker previously used. The filter paper with the acid insoluble residue shall be saved for the determination of acid insoluble material as specified in 4.3.3. Add to the beaker 3 ml of 15 normal (N.) nitric acid which has been freshly boiled to remove the oxides of nitrogen. Electrolyze the solution at room temperature with a current of 3.0 amperes using a tared platinum gauze cathode and rotating platinum anode. Add 1 gm. of urea to the beaker when the blue color of the solution has been discharged and continue the electrolysis for an additional 15 minutes. Test for completeness of electrolysis by adding about 20 ml of water, electrolyzing for an additional 15 minutes and observing if any copper deposited on the upper part of the stem of the cathode. If electrolysis is not complete continue until it is complete as indicated by this procedure. Wash the apparatus with water while the current is still flowing, remove the cathode from the apparatus, dip in acetone, dry in air and weigh.

Calculate the percent cupric oxide in the sample as follows:

$$\text{Percent cupric oxide} = \frac{125.18A}{W}$$

Where:

A = Weight of copper deposited.

W = Weight of sample corrected for moisture.

4.3.3 *Determination of acid insoluble material.* The filter paper holding the residue remaining after the wash treatment specified in 4.3.2 shall be transferred to a tared crucible and ignited over a flame. The crucible and residue shall be cooled in a desiccator and weighed. The increase in weight shall be calculated to percentage of acid insoluble material.

4.3.4 *Granulation.* A bottom pan shall be attached to a 74 micron (No. 200) sieve, as specified in Specification RR–S–366. A weighed 100 gm. portion of the sample shall be placed on the sieve. The sample shall be covered and shaken for 5 minutes by means of a mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Any material remaining on the sieve shall be weighed and the percentage which has passed through the sieve calculated.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packing, for levels A, B and C. Not applicable.

5.2 Packing (see 6.2).

5.2.1 *Level A.* Fifty pounds of cupric oxide shall be packed in a shipping sack conforming to type II, construction No. 2X, Specification UU–S–48. The closure and waterproofing of the ends shall be accomplished in accordance with the requirements of Specification UU–S–48.

5.2.2 *Level C.* Cupric oxide shall be packed for shipment in conformance with consolidated freight classification rules and container specifications for rail shipments, or with National Motor Freight classification rules and container specifications for truck shipments.

5.3 Marking. Containers shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 Intended use. This specification covers cupric oxide for use in pyrotechnic compositions.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of specification.

(b) Levels of protection (see 5.2).

Notice. When Government drawings, specifications, or other data are used for any purpose other than in

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:54Z
Check the source to verify that this is the current version before use.

connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may be related thereto.

Custodian:
Army—Ord
Navy—WEP
Air Force—MAA

Preparing activity:
Army—Ord

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:54Z
Check the source to verify that this is the current version before use.